

MATHEMATICAL REASONING

1. The equation for the statement 'One-third of a number added to second multiple of 6 is 24' is _____

- (A) $y/3 = 12 + 24$ (B) $y/3 + 2 = 24$ (C) $y/3 + 12 = 24$ (D) $y/3 = 18 + 24$

2. When Raju multiplies a certain number by 17 and adds 4 to the product, he gets 225. Find the number.

- (A) 13 (B) 14 (C) 15 (D) 16

3. Equation for the statement 'Twice the product of m and n is equal to thrice their difference' is _____

- (A) $3mn = 2(m - n)$ (B) $mn = 2(m - n)$ (C) $2mn = m - n$ (D) $2mn = 3(m - n)$

4. If six times of a number is 318, then the number is _____

- (A) 53 (B) 91 (C) 0 (D) 72

5. An algebraic expression, $11 - y$ can be written in statement form as _____

- (A) 11 less than y (B) y less than 11 (C) y more than 11 (D) y divided by 11

6. Which of the following equations justify the given statement? "When x is divided by y, the quotient is added to the product of x and y"

- (A) $x + y/x$ (B) $y/x + yx$ (C) $yx + 2x/y$ (D) $x/y + xy$

7. If twelve less than 3 times of a number is 27, then the number is _____

- (A) 29 (B) 39 (C) 65 (D) 13

8. If $x = 3$ and $y = 7$, then find the value of $x/2 + 4y + 2$.

- (A) $63/2$ (B) $65/2$ (C) $17/2$ (D) 96

9. To get the value of 'x', the number to be multiplied on either side of equation $x/5 = 15/3$ is _____

- (A) -3 (B) 15 (C) 5 (D) 0

10. If $(2/3)p - 2(1/2) = 3(1/2)$, then the value of p is _____

- (A) -9 (B) 6 (C) 9 (D) 0

11. Which of the following is an equation?

- (A) $n - 11$ (B) $n + 1 > 0$ (C) $n + 1 = 0$ (D) $11 - n$

12. The breadth of a rectangle is w cm and the length is 5 times as long as its breadth. What is the perimeter of the rectangle?

- (A) $5w$ cm (B) $12w$ cm (C) $(10 + 2w)$ cm (D) $(25 + w)$ cm

13. The number of girls in a class is 3 times the number of boys. Which of the following cannot be the total number of students in the class?

- (A) 24 (B) 32 (C) 36 (D) 41

14. The algebraic expression for the statement "Product of x and reciprocal of a is subtracted from the product of y and reciprocal of b" is _____

- (A) $y/b - x/a$ (B) $(y - x)/(a - b)$ (C) $xa - yb$ (D) $1/(yb - xa)$

than 2 times the number x can be written in algebraic form as **Q**. (iii) An equation is a condition on a **R**.

Option	P	Q	R
(A)	variable	$8x + 2$	equation
(B)	equation	$2x + 8$	variable
(C)	equation	$2x - 8$	variable
(D)	variable	$2x + 8$	equation

28. Which of the following equations does not have a solution in integers?

(A) $x + 1 = 1$ (B) $x - 1 = 3$ (C) $2x + 1 = 6$ (D) $1 - x = 5$

29. State 'T' for true and 'F' for false.

P. $x = 15$ is the solution of the equation $41 - x = 25$.

Q. An equation is an algebraic expression which involves an "equal to" sign.

R. ' x exceeds y by 7' can be expressed as $x = y + 7$.

Option	P	Q	R
(A) F, T, T	(B) F, T, F	(C) T, F, T	(D) T, T, T

30. Match the following.

Column-I	Column-II
(i) The total weight of 3 boxes is 5 kg. If the weight of two of the boxes is x kg each, then the weight of third box (in kg) is	(a) $x - 11$
(ii) Sid had x toffees. He ate 5 toffees and gave 6 toffees to his neighbour. The number of toffees left with him is	(b) $x/3$
(iii) Mohit had Rs x . He gave the money to his 3 sisters equally. Each girl will get Rs	(c) $5 - 2x$

Options:

(A) (i) $\rightarrow$ (c), (ii) $\rightarrow$ (a), (iii) $\rightarrow$ (b) (B) (i) $\rightarrow$ (b), (ii) $\rightarrow$ (c), (iii) $\rightarrow$ (a) (C) (i) $\rightarrow$ (c), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (a) (D) (i) $\rightarrow$ (a), (ii) $\rightarrow$ (b), (iii) $\rightarrow$ (c)